Estimation Theory

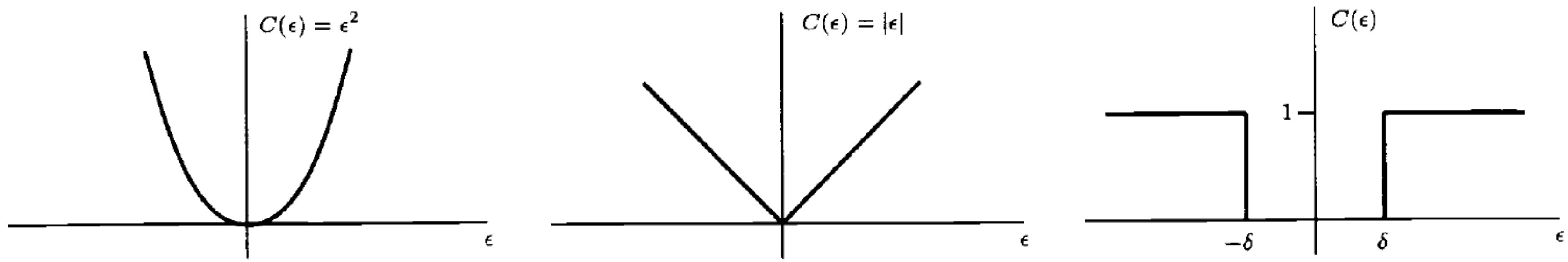
General Bayesian Estimators

Bayes Risk

$$\boxed{\mathcal{R} = E[\mathcal{C}(\epsilon)]} \qquad \epsilon = \theta - \hat{\theta}$$

Error.

Various Cost Function



(a) Quadratic error **(b)** Absolute error **(c)** Hit-or-miss error

$$\mathcal{C}(\epsilon) = \begin{cases} 0 & |\epsilon| < \delta \\ 1 & |\epsilon| > \delta \end{cases}$$

threshold.

$$\begin{aligned} \mathcal{R} &= E[\mathcal{C}(\epsilon)] \\ &= \int\int \mathcal{C}(\theta - \hat{\theta}) p(\mathbf{x}, \theta)\, d\mathbf{x}\, d\theta \\ &= \int \left[\int \mathcal{C}(\theta - \hat{\theta}) p(\theta|\mathbf{x})\, d\theta \right] p(\mathbf{x})\, d\mathbf{x} \end{aligned}$$

Bayes rule

Bayes Risk

Quadratic error: $C(e)=e^2$

MMSE estimator: $\hat{\theta} = E(\theta|\mathbf{x})$

Absolute error: $C(e)=|e|$

$$
\begin{aligned}
g(\hat{\theta}) &= \int |\theta - \hat{\theta}| p(\theta|\mathbf{x})\, d\theta \\
&= \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) p(\theta|\mathbf{x})\, d\theta + \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) p(\theta|\mathbf{x})\, d\theta.
\end{aligned}
$$

$$
\frac{dg(\hat{\theta})}{d\hat{\theta}} = \int_{-\infty}^{\hat{\theta}} p(\theta|\mathbf{x})\, d\theta - \int_{\hat{\theta}}^{\infty} p(\theta|\mathbf{x})\, d\theta = 0
$$

$$
\int_{-\infty}^{\hat{\theta}} p(\theta|\mathbf{x})\, d\theta = \int_{\hat{\theta}}^{\infty} p(\theta|\mathbf{x})\, d\theta
$$

$\hat{\theta}$ 이 mid point
= median

$\hat{\theta}$ is the *median* of the posterior PDF

Hit-or-miss error:

$\hat{\theta}-\delta \sim \hat{\theta}+\delta$ 는 0 이로

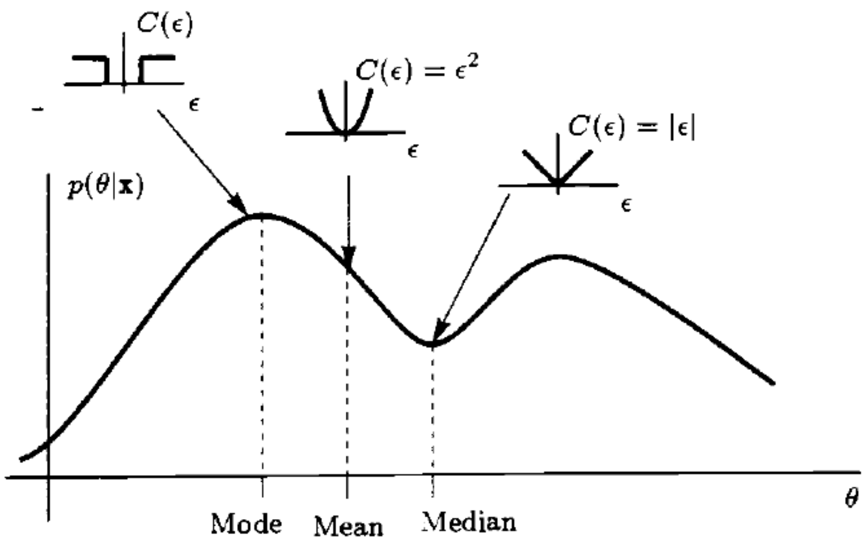
$$
g(\hat{\theta}) = \int_{-\infty}^{\hat{\theta}-\delta} 1 \cdot p(\theta|\mathbf{x})\, d\theta + \int_{\hat{\theta}+\delta}^{\infty} 1 \cdot p(\theta|\mathbf{x})\, d\theta
$$

$$
g(\hat{\theta}) = 1 - \int_{\hat{\theta}-\delta}^{\hat{\theta}+\delta} p(\theta|\mathbf{x})\, d\theta
$$

$[\hat{\theta}-\delta,\ \hat{\theta}+\delta]$

For sufficiently small δ, $\hat{\theta} = \arg\max p(\theta|\mathbf{x})$, i.e., the *mode* of the posterior PDF. This is known as the *maximum a posteriori* (MAP) estimator.

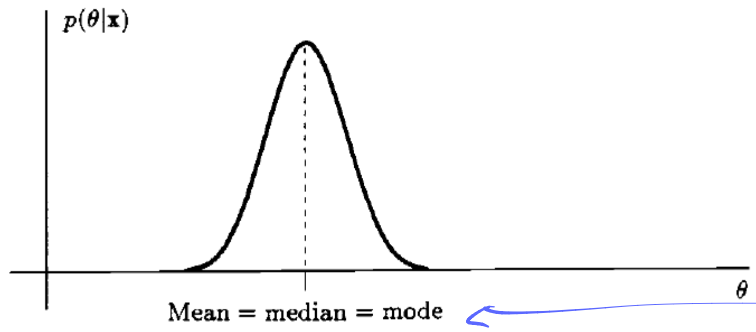
Bayes Risk



(a) General posterior PDF

	Quadratic error	Absolute error	Hit-or-miss error
Posterior's	Mean	Median	Mode

Gaussian: $$p(\theta|\mathbf{x}) = \frac{1}{\sqrt{2\pi\sigma^2_{\theta|x}}} \exp\left[-\frac{1}{2\sigma^2_{\theta|x}}(\theta - \mu_{\theta|x})^2\right]$$



(b) Gaussian posterior PDF

posterior Distribution이 Gaussian일 때는

<u>Bayesian Linear Model</u>

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \qquad \theta \sim \mathcal{N}(\mu_\theta, \mathbf{C}_\theta), \mathbf{w} \sim \mathcal{N}(0, \mathbf{C}_w)$$

$$E(\boldsymbol{\theta}|\mathbf{x}) = \boldsymbol{\mu}_\theta + \mathbf{C}_\theta \mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta) \;= \boldsymbol{\mu}_\theta + \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta)$$

$$\mathbf{C}_{\theta|x} = \mathbf{C}_\theta - \mathbf{C}_\theta\mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}\mathbf{H}\mathbf{C}_\theta \;= (\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}$$

Matrix inversion lemma:

$$\begin{aligned}(A_{11} + A_{12}A_{22}^{-1}A_{21})^{-1} &= A_{11}^{-1} - A_{11}^{-1}A_{12}(A_{22} + A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} \\ (A_{11} + A_{12}A_{22}^{-1}A_{21})^{-1}A_{12}A_{22}^{-1} &= A_{11}^{-1}A_{12}(A_{22} + A_{21}A_{11}^{-1}A_{12})^{-1}\end{aligned}$$

$$\mathbf{z} = [\mathbf{x}^T\ \boldsymbol{\theta}^T]^T$$

$$\mathbf{z} = \begin{bmatrix}\mathbf{H}\boldsymbol{\theta} + \mathbf{w} \\ \boldsymbol{\theta}\end{bmatrix} = \begin{bmatrix}\mathbf{H} & \mathbf{I} \\ \mathbf{I} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\boldsymbol{\theta} \\ \mathbf{w}\end{bmatrix}$$

$$\begin{aligned}E(\mathbf{x}) &= E(\mathbf{H}\boldsymbol{\theta} + \mathbf{w}) = \mathbf{H}E(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\mu}_\theta \\ E(\mathbf{y}) &= E(\boldsymbol{\theta}) = \boldsymbol{\mu}_\theta\end{aligned}$$

$$\begin{aligned}\mathbf{C}_{xx} &= E\left[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\right] \\ &= E\left[(\mathbf{H}\boldsymbol{\theta} + \mathbf{w} - \mathbf{H}\boldsymbol{\mu}_\theta)(\mathbf{H}\boldsymbol{\theta} + \mathbf{w} - \mathbf{H}\boldsymbol{\mu}_\theta)^T\right] \\ &= E\left[(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})^T\right] \\ &= \mathbf{H}\underbrace{E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)^T\right]}_{C_\theta}\mathbf{H}^T + \underbrace{E(\mathbf{w}\mathbf{w}^T)}_{C_w} \\ &= \mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w\end{aligned}$$

$$\begin{aligned}\mathbf{C}_{yx} &= E\left[(\mathbf{y} - E(\mathbf{y}))(\mathbf{x} - E(\mathbf{x}))^T\right] \\ &= E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})^T\right] \\ &= E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta))^T\right] \\ &= \mathbf{C}_\theta\mathbf{H}^T.\end{aligned}$$

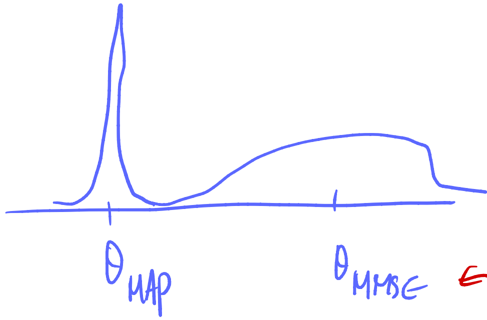
In the absence of prior knowledge, $\Rightarrow \mu_\theta = 0$, $C_\theta \to \infty$ ← prior information이 없다는 것의 의미

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta}|\mathbf{x}) = \cancelto{0}{\boldsymbol{\mu}_\theta} + (\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}\cancelto{0}{\boldsymbol{\mu}_\theta})$$

$$\mathbf{C}_\theta^{-1} \to \mathbf{0} \qquad \hat{\boldsymbol{\theta}} \to (\mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{x} \qquad \text{(MVU estimator)}$$

Maximum A Posteriori (MAP) Estimators

$$\boxed{\hat{\theta} = \arg\max_{\theta} p(\theta|\mathbf{x})} \qquad p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{p(\mathbf{x})}$$



← Better Estimator!

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} p(\theta|\mathbf{x}) \\ &= \arg\max_{\theta} p(\mathbf{x}|\theta)p(\theta) \\ &= \arg\max_{\theta}[\log p(\mathbf{x}|\theta) + \log p(\theta)]\end{aligned}$$

- Easy to compute (cf. MMSE)
- Not necessarily the best for all cases

Example: Exponential PDF

$$p(x[n]|\theta) = \begin{cases} \theta \exp(-\theta x[n]) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases} \qquad p(\theta) = \begin{cases} \lambda \exp(-\lambda\theta) & \theta > 0 \\ 0 & \theta < 0 \end{cases}$$

For N i.i.d. data points, $p(\mathbf{x}|\theta) = \prod_{n=0}^{N-1} p(x[n]|\theta)$

$$\begin{aligned} g(\theta) &= \ln p(\mathbf{x}|\theta) + \ln p(\theta) \\ &= \ln\left[\theta^N \exp\left(-\theta \underbrace{\sum_{n=0}^{N-1} x[n]}_{N\bar{x}}\right)\right] + \ln[\lambda \exp(-\lambda\theta)] \\ &= N\ln\theta - N\theta\bar{x} + \ln\lambda - \lambda\theta \end{aligned}$$

$$\frac{dg(\theta)}{d\theta} = \frac{N}{\theta} - N\bar{x} - \lambda$$

$= 0$

MAP estimator: $\hat{\theta} = \dfrac{1}{\bar{x} + \frac{\lambda}{N}}$

As $N \to \infty$, $\hat{\theta} \to 1/\bar{x}$.
As $\lambda \to 0$, $\hat{\theta} \to 1/\bar{x}$ (MLE).

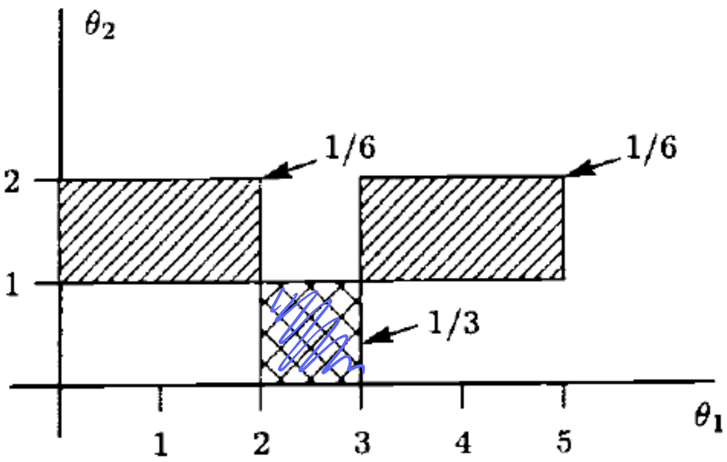
Scalar MAP Estimator vs. Vector MAP Estimator

Scalar MAP estimator: $\hat{\theta}_i = \arg\max_{\theta_i} p(\theta_i|\mathbf{x}) \qquad i = 1, 2, \ldots, p.$

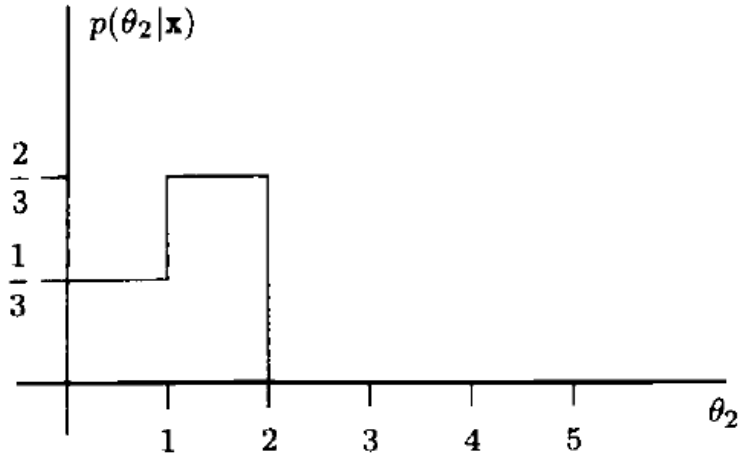
$$p(\theta_1|\mathbf{x}) = \int \cdots \int p(\boldsymbol{\theta}|\mathbf{x})\, d\theta_2 \ldots d\theta_p$$

marginalize all other parameters.

Vector MAP estimator: $\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\mathbf{x})$



(a) Posterior PDF $p(\theta_1, \theta_2|\mathbf{x})$



(b) Posterior PDF $p(\theta_2|\mathbf{x})$

MAP estimator: $2 < \hat{\theta}_1 < 3, 0 < \hat{\theta}_2 < 1$ $\neq$ MAP estimator: $1 < \hat{\theta}_2 < 2$

Not the same !!!

Invariance Properties of the MLE

Recall

Theorem 7.2 (Invariance Property of the MLE) *The MLE of the parameter $\alpha = g(\theta)$, where the PDF $p(\mathbf{x};\theta)$ is parameterized by θ, is given by*

$$\hat{\alpha} = g(\hat{\theta}) \quad \checkmark$$

where $\hat{\theta}$ is the MLE of θ. The MLE of $\hat{\theta}$ is obtained by maximizing $p(\mathbf{x};\theta)$. If g is not a one-to-one function, then $\hat{\alpha}$ maximizes the modified likelihood function $\bar{p}_T(\mathbf{x};\alpha)$, defined as

$$\bar{p}_T(\mathbf{x};\alpha) = \max_{\{\theta:\alpha=g(\theta)\}} p(\mathbf{x};\theta).$$

$$x[n] = A + w[n], \quad n = 0, 1, \ldots, N-1, \quad \text{where } w[n] \sim \mathcal{N}(0,\sigma^2)$$

When $\alpha = \exp(A)$, find the MLE of α.

$$p(\mathbf{x};A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right] \qquad -\infty < A < \infty$$

$$p_T(\mathbf{x};\alpha) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-\ln\alpha)^2\right] \qquad \alpha > 0$$

$$\ldots\ldots \quad \sum_{n=0}^{N-1}(x[n]-\ln\hat{\alpha})\frac{1}{\hat{\alpha}} = 0 \qquad \hat{\alpha} = \exp(\bar{x}) \qquad \hat{\alpha} = \exp(\hat{A}) = \exp(\hat{\theta})$$

Invariance Property Does Not Hold for the MAP Estimator

Example: Exponential PDF

$$p(x[n]|\theta) = \begin{cases} \theta \exp(-\theta x[n]) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases} \qquad p(\theta) = \begin{cases} \lambda \exp(-\lambda\theta) & \theta > 0 \\ 0 & \theta < 0 \end{cases}$$

For N iid data pts, $p(\mathbf{x}|\theta) = \prod_{n=0}^{N-1} p(x[n]|\theta)$

MAP estimator: $\hat{\theta} = \dfrac{1}{\bar{x} + \frac{\lambda}{N}}$

Suppose we want to estimate $\alpha = 1/\theta$. $\quad \hat{\alpha} = \dfrac{1}{\hat{\theta}} \quad \hat{\alpha} = \bar{x} + \dfrac{\lambda}{N}$???

$$\underbrace{p_\alpha(\alpha)}_{\text{density}} \cdot \underbrace{|d\alpha|}_{\text{unit length}} = p_\theta(\theta(\alpha)) \cdot |d\theta|$$

$$p(x[n]|\alpha) = \begin{cases} \frac{1}{\alpha} \exp\left(-\frac{x[n]}{\alpha}\right) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases}$$

$$p_\alpha(\alpha) = \frac{p_\theta(\theta(\alpha))}{\left|\frac{d\alpha}{d\theta}\right|} = \begin{cases} \frac{\lambda \exp(-\lambda/\alpha)}{\alpha^2} & \alpha > 0 \\ 0 & \alpha < 0 \end{cases}$$

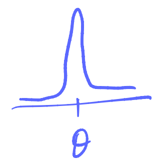
$$\begin{aligned} g(\alpha) &= \ln p(\mathbf{x}|\alpha) + \ln p(\alpha) \\ &= \ln\left[\left(\frac{1}{\alpha}\right)^N \exp\left(-\frac{1}{\alpha}\sum_{n=0}^{N-1} x[n]\right)\right] + \ln \frac{\lambda \exp(-\lambda/\alpha)}{\alpha^2} \\ &= -N\ln\alpha - N\frac{\bar{x}}{\alpha} + \ln\lambda - \frac{\lambda}{\alpha} - 2\ln\alpha \\ &= -(N+2)\ln\alpha - \frac{N\bar{x}+\lambda}{\alpha} + \ln\lambda. \end{aligned}$$

$$\frac{dg}{d\alpha} = -\frac{N+2}{\alpha} + \frac{N\bar{x}+\lambda}{\alpha^2} = 0$$

MAP estimator: $\hat{\alpha} = \dfrac{N\bar{x}+\lambda}{N+2}$

Performance Description

- $p(\hat{\theta}|\theta)$, different PDF of the estimator for each realization of θ.
- Good estimator: close to θ for every possible value of θ, i.e., $\epsilon = \theta - \hat{\theta}$ is small.
- Good estimator $\hat{\theta}$: $p(\hat{\theta}|\theta)$ should be concentrated about θ for all possible θ.
- Such estimator is the MMSE estimator $\hat{\theta} = \mathbb{E}(\theta|\mathbf{x})$, which minimizes $\mathbb{E}_{x,\theta}((\theta - \hat{\theta})^2)$.



$$\epsilon = \theta - E(\theta|\mathbf{x})$$

$$\begin{aligned} E_{x,\theta}(\epsilon) &= E_{x,\theta}\left[\theta - E(\theta|\mathbf{x})\right] \\ &= E_x\left[E_{\theta|x}(\theta) - E_{\theta|x}(\theta|\mathbf{x})\right] \\ &= E_x\left[E(\theta|\mathbf{x}) - E(\theta|\mathbf{x})\right] = 0 \checkmark \end{aligned}$$

$$\begin{aligned} \mathrm{var}(\epsilon) &= E_{x,\theta}(\epsilon^2) \\ &= E_{x,\theta}\left[(\theta - \hat{\theta})^2\right] \end{aligned}$$

Bayesian MSE

if ϵ is Gaussian, we have that $\epsilon \sim \mathcal{N}(0, \mathrm{Bmse}(\hat{\theta}))$

DC level in WGN – Gaussian prior

$$\hat{A} = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\bar{x} + \frac{\frac{\sigma^2}{N}}{\sigma_A^2 + \frac{\sigma^2}{N}}\mu_A$$

$$\epsilon \sim \mathcal{N}\left(0, \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}\right)$$

$$\mathrm{Bmse}(\hat{A}) = \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}$$

Consistent in the Bayesian sense, i.e., for large data the estimator will always be close to the realization of the parameter.